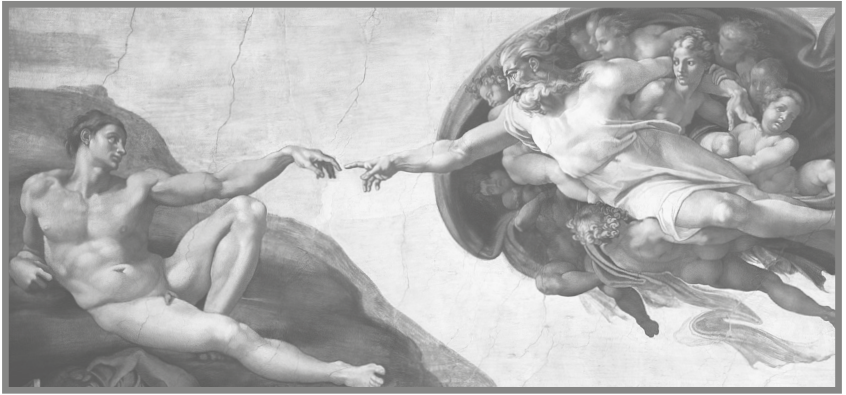




the building began in 1296 no one had figured out how to make a dome to cover it. The dome would have to be the largest cupola constructed since the late Roman period and the guild specified that it should be built without external buttresses, favored by their political rivals in France, Germany, and Milan and also considered old-fashioned. This seemed an impossible task. The young goldsmith and clockmaker-turned-architect Filippo Brunelleschi won the competition with his daring plan for a huge eight-sided brick dome, but many doubted that he would be able to construct it.

The main problem was being able to support the structure in such a way that it did not spread and collapse under its own weight. Brunelleschi's ingenious solution was to construct two concentric domes—an inner supporting dome and a larger outer one. The domes were then joined together with huge brick arches and a complex interlocking system of “chains” made from rings of stone and wooden beams that were attached by iron clamps to prevent the dome from expanding outward.

The result—which was finally completed in 1436—remains the largest masonry dome in the world. Combining the style of antiquity with new engineering techniques, it exhibited the blend of ancient wisdom and modern knowledge that typified the Renaissance.

Michelangelo's painted ceiling at the Sistine Chapel in the Vatican combines the Renaissance interest in physical beauty and realism with religious subject matter.

The Renaissance in Italy

Meaning “rebirth,” the Renaissance was a movement that started in Italy and began to spread across Europe from the mid-15th century. Its roots lay in the rediscovery of the culture of ancient Greece and Rome and it influenced all the arts, as well as science and scholarship. Painters, sculptors, and architects broke free from the traditions of medieval art. They visited the monuments of ancient Rome, looking at classical statues and the carvings on Roman buildings, and created works of art in the classical style. This new movement inspired architects, such as Leon Battista Alberti and Brunelleschi, and a wave of great artists, including Michelangelo and Leonardo da Vinci. Most of these figures were active in many fields—Brunelleschi was a sculptor and engineer as well as an architect; Michelangelo painted, sculpted, and wrote poetry; while da Vinci's achievements spanned both the arts and the sciences.

Renaissance painters and sculptors sought to represent the physical world in a more realistic way than their Medieval predecessors: they valued anatomical accuracy and developed scientific methods of illustrating perspective. As in classical art, there was more focus on human beauty and the nude. »